Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3471-WE01

Title: MEng and BEng Engineering

Materials 3

Time Allowed:	2 hours	
Additional Material provided:	Formula sheets at the end of the paper.	
Materials Permitted:		
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. Section A is worth 60%. Section B is worth 40%. Questions within a section carry equal weight. All relevant workings must be shown.

Revision:	

SECTION A

Question 1

a) What is a Frank-Read source? Explain the mechanism by illustrating the dislocation line movement.

[5%]

b) Determine the Miller indices for the planes A, B & C in the cubic unit cell shown in figure Q1a.

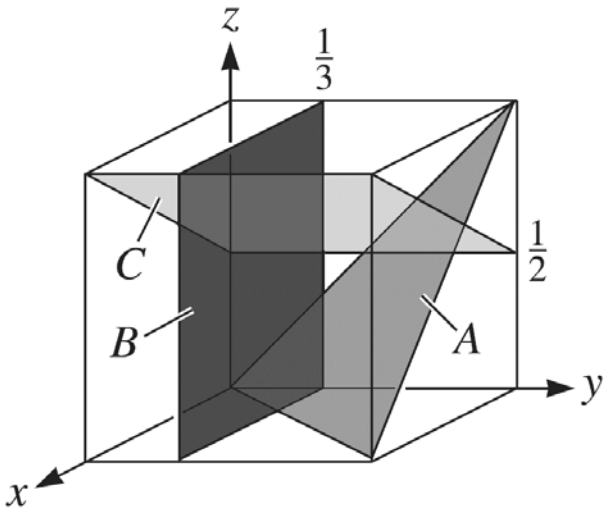


Figure Q1a

[15%]

c) In a Brass (Cu-Zn) alloy, Zn atoms are located at sites such as (0, 0, ½). The lattice parameter is 0.42 nm for FCC Cu. Assume that Zn atoms have a radius of 0.14nm.

i. What would be the ratio of the interstitial site to the radius of the zinc?

ii. What would be the atomic percentage of zinc in FCC copper if all the interstitial sites were filled?

[40%]

d) A copper-zinc alloy has the following properties:

Grain Diameter (mm)	Strength (MPa)
0.0009	1700
0.0025	1580

Determine the grain size required to obtain strength of 2000 MPa. **[10%]**

e) We want a copper bar (circular cross section) to have a tensile strength of at least 550 MPa and a final diameter of 9.525mm after cold working. What is the minimum diameter of the original bar? (See Figure Q1b.)

[10%]

continued

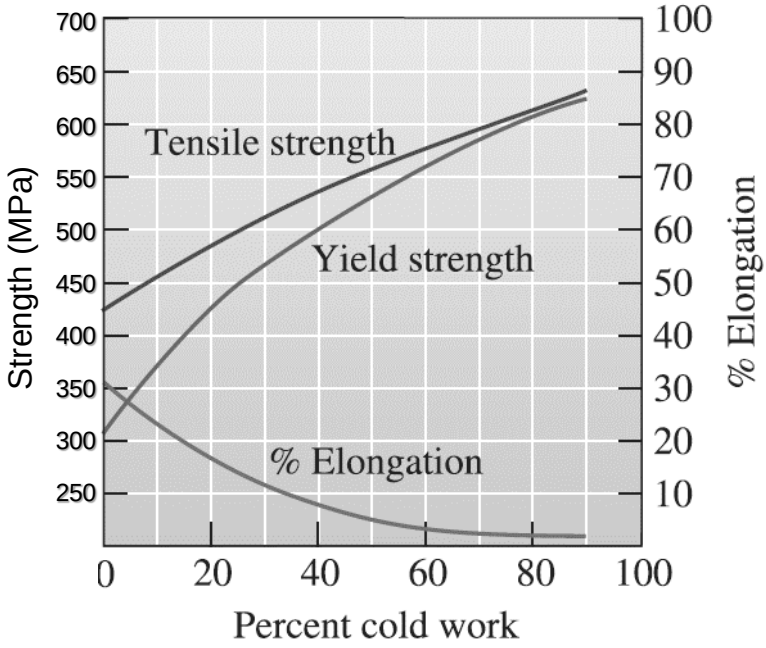


Figure Q1b

f) A 10 kg pipe made of magnesium 80%-aluminium (see phase diagram – Figure Q1c) alloy is used for transferring a corrosive material. The pipe dimension is as follows: inside-diameter is 10cm, its wall thickness is 0.38 cm and it is 400 cm-long. The pipe is accidentally connected to the power system of a manufacturing plant at room temperature, causing a current of 0.05A to flow through the pipe. If γ is a perfect corrosion-resistant material, estimate the time required before the pipe begins to leak. Assume a uniform rate of corrosion.

Note that Faraday's constant is 96500 C, n=3, $\rho_\alpha = 1.5\ \frac{g}{cm^3}$, $\rho_\gamma = 2.5\ \frac{g}{cm^3}$, $M_{\alpha,RT}$ = 40 g/mol, and $M_{\gamma,RT}$ = 50 g/mol.

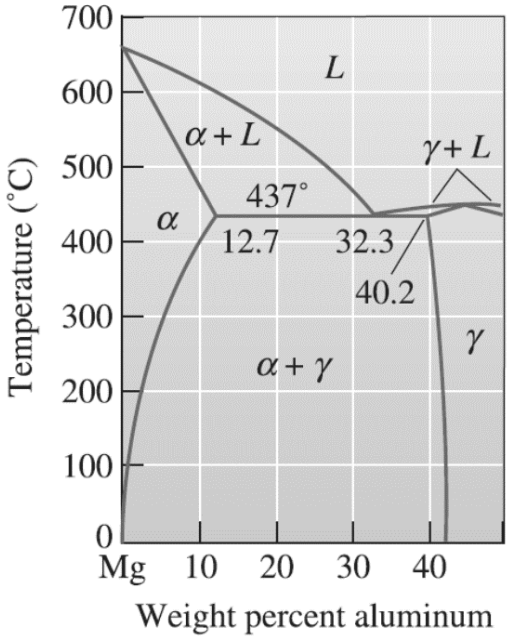


Figure Q1c – Phase Diagram

[20%]

continued

Question 2

a) Silicon carbide (SiC) has a specific gravity of 3.1 g/cm^3. A sintered SiC part is produced, occupying a volume of 500 cm^3 and weighing 1200 g. After soaking in water, the part weighs 1250 g. Calculate the true porosity and the volume fraction of the total porosity that consists of closed pores.

[30%]

b) Name 5 parameters affect the "Thermal Shock Resistance" of materials. **[15%]**

c) Name 4 types of Adhesives. **[10%]**

d) Define (i) a thermoplastic, (ii) thermosetting plastics, and (iii) elastomers. **[15%]**

e) Glass fibres in nylon provide reinforcement. If the nylon contains 30 vol% E-glass, what fraction of the force applied parallel to the fibre axis is carried by the glass fibres? The modulus of elasticity for each component of the composite is:

$$E_{glass} = 10.5 \times 10^6 psi, \quad E_{nylon} = 0.4 \times 10^6 psi$$

[30%]

continued

SECTION B

Question 3

a) What is sustainable development and why is it relevant to engineers? **[10%]**

b) What are the two main models for achieving sustainable development? **[10%]**

c) Discuss the pros and cons of each model economically, socially and environmentally. **[30%]**

d) Explain the difference between int**ra**generational and int**er**generational equity by giving an example of each by with respect to one Sustainable Development Goal. **[15%]**

e) Use a diagram to help explain how the Kyoto Protocol is economically, socially and environmentally relevant to int**er** and int**ra** generational equity. **[35%]**

Question 4

You are required to select a material for a cantilever axle for a racing car. The axle must not fail and should be as light as possible. Raw material cost is not a primary consideration, but the part will need to be manufacturable to a suitable tolerance to mate with wheel bearings etc. You can initially assume that there is a free choice of diameter(s) for the axle and that other components can be sized to suit the axle. The design maximum bending moment in the axle from the external loading of the wheel is 1250 Nm (incorporating an appropriate factor of safety) and occurs at the cantilever root where the axle meets the outboard wheel bearing.

Figure Q4a illustrates the axle, and outboard wheel bearing. Key properties for a range of materials are provided in Table QT4a and Figure Q4b. Table QT4b provides material indices for typical applications.

a) What is the appropriate loading case and material index in this case? Sketch a materials selection chart with a suitable screening line. The sketch chart only needs to show the axes and screening line; you do not need to plot points for any materials and it does not need to be to scale.

[25%]

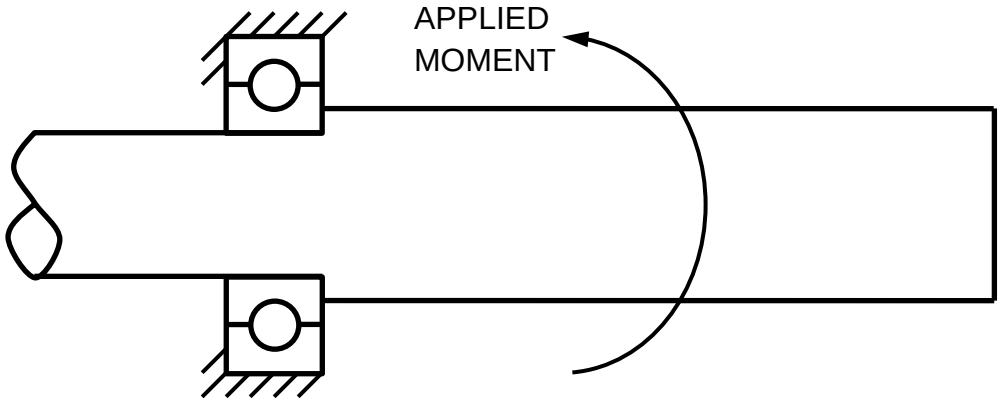
b) List six materials which appear to be good choices according to the material index. **[25%]**

c) An additional constraint is imposed that the axle diameter, at the point of peak bending moment where it meets the wheel bearing, must not exceed 40 mm. Illustrate how this constraint could be included on your material selection chart sketch. Your illustration does not need to be to scale, but, if possible, include the equation for the constraint line on the chart.

[30%]

d) Given the constraint in c) and any other considerations recommend a material for this application.

[20%]



Q4a – Axle and Bearing Assembly – Not to Scale

continued

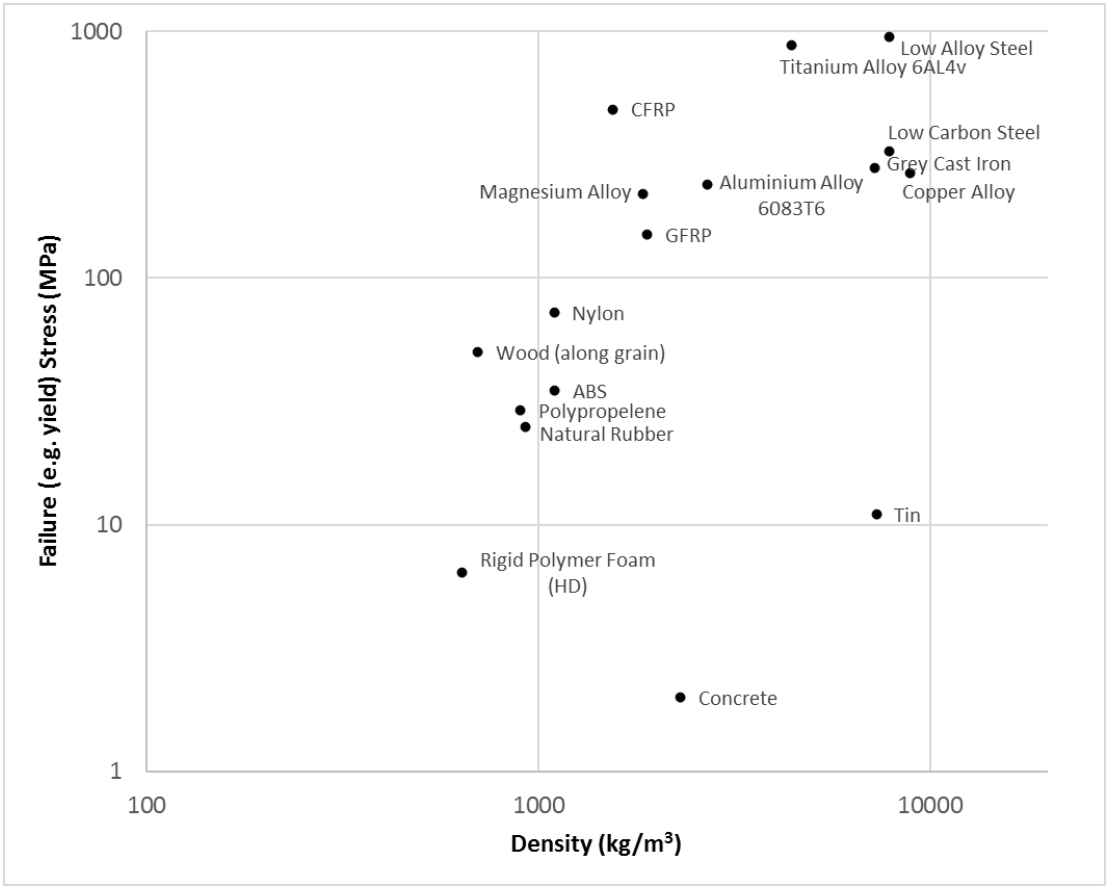


Figure Q4b – Materials Selection Chart

Material	Density (kg/m^3)	Young's Modulus (GPa)	Failure Stress (MPa)	$\frac{E}{\rho} \times 10^6$ m^2s^{-2}	$\frac{E^{1/2}}{\rho}$ $m^{\frac{5}{2}}s^{-1}kg^{-\frac{1}{2}}$	$\frac{\sigma_f}{\rho} \times 10^3$ m^2s^{-2}	$\frac{\sigma_f^{2/3}}{\rho}$ $m^{\frac{7}{3}}s^{-\frac{4}{3}}kg^{-\frac{1}{3}}$
Copper Alloy	8900	130	265	14.6	40.5	29.8	46
Low Alloy Steel	7850	215	950	27.4	59.1	121.0	123
Low Carbon Steel	7850	210	325	26.8	58.4	41.4	60
Tin	7300	43	11	5.9	28.4	1.5	7
Grey Cast Iron	7200	110	280	15.3	46.1	38.9	59
Titanium Alloy	4430	113.5	880	25.6	76.0	198.6	207
Aluminium Alloy	2700	70	240	25.9	98.0	88.9	143
Concrete	2300	20	2	8.7	61.5	0.9	7
GFRP	1900	21.5	150	11.3	77.2	78.9	149
Magnesium Alloy	1850	44.5	220	24.1	114.0	118.9	197
CFRP	1550	55.1	483	35.5	151.4	311.6	397
Nylon	1100	2.9	72.5	2.6	49.0	65.9	158
ABS	1100	2	35	1.8	40.7	31.8	97
Natural Rubber	925	0.002	25	0.002	1.5	27.0	92
Polypropylene	900	1.25	29	1.4	39.3	32.2	105
Wood (along grain)	700	13	50	18.6	162.9	71.4	194
Rigid Polymer Foam (HD)	640	0.34	6.4	0.5	28.8	10.0	54

Table QT4a – Material Properties and Indices

continued

	Stiff	Strong
Ties	$\frac{E}{\rho}$	$\frac{\sigma_f}{\rho}$
Beams	$\frac{E^{1/2}}{\rho}$	$\frac{\sigma_f^{2/3}}{\rho}$
Shafts	$\frac{E^{1/2}}{\rho}$	$\frac{\sigma_f^{2/3}}{\rho}$
Plates	$\frac{E^{1/3}}{\rho}$	$\frac{\sigma_f^{1/2}}{\rho}$

Table QT4b – Material Indices for Standard Cases

Question 5

a) A fillet weld has a cross-sectional area of 20 mm^2 and is 400mm long.

i) What quantity of heat (in joules) is required to accomplish the weld if the metal to be welded is high carbon steel? K = 3.33 x 10^{-6} $JK^{-2}mm^{-3}$, melting temperature (T_m) for high carbon steel = 1650 °K.

[40%]

ii) How much heat must be generated at the welding source if the heat transfer factor is 0.85 and the melting factor is 0.68?

[20%]

b) Describe with the aid of diagrams the steps in creating a Rapid Prototype model.

[40%]

continued

Formula Sheet - Metals-Ceramics- Polymers- Composites

Linear Thermal Expansion	$\alpha = \left(\frac{1}{L}\right)\left(\frac{dL}{dT}\right)$
ASTM Grain Size number	$N = 2^{n-1}$
Hall-Petch equation	$\sigma_y = \sigma_0 + kd^{-1/2}$
Cold work	$Percent\ reduction\ in\ thickness = (\frac{t_0 - t_f}{t_0}) \times 100$
Cold work	$Percent\ of\ cold\ work = (\frac{A_0 - A_f}{A_0}) \times 100$
Faraday's Equation	$w = \frac{ItM}{nF}$
Apparent Porosity	$\text{Apparent Porosity} = \frac{W_W - W_d}{W_W - W_s} \times 100$
True Porosity	$\text{True Porosity} = \frac{\rho - B}{\rho} \times 100$
Bulk Density	$\text{B} = \frac{W_d}{W_W - W_s}$
Maxwell Stress	$s_z = \frac{-\varepsilon\varepsilon_0 V^2}{E}$
Rule of mixtures	$\boldsymbol{\rho_c = \sum(f_i \rho_i) = f_1\rho_1 + f_2\rho_2 + \ \ldots + f_n\rho_n}$
rule of mixtures for the modulus of elasticity of a fibre reinforced composite when stress (σ) is applied along the axis of the fibres.	$\boldsymbol{E_{c,\parallel} = f_m E_m + f_f E_f}$
rule of mixtures for the modulus of elasticity of a fibre reinforced composite when stress (σ) is applied perpendicular to the axis of the fibres.	$\boldsymbol{\frac{1}{E_{c,\perp}} = \frac{f_m}{E_m} + \frac{f_f}{E_f}}$

continued

Formula sheet - Material Processing

Screen mesh for sorting particle sizes

$$PS = \frac{1}{MC} - t_w$$

Where PS = particle size, MC = mesh count, openings per linear mm and t_w = wire thickness of screen mesh.

Surface area (sphere)

$$A = \pi D^2$$

Where D = diameter of the spherical particle

Surface volume (sphere)

$$V = \frac{\pi D^3}{6}$$

Where D = diameter of the spherical particle

Area-to-volume ratio for a sphere

$$\frac{A}{V} = \frac{6}{D}$$

Where A = surface area, V = surface volume, D = diameter of the spherical particle

Area-to-volume ratio for any particle shape – spherical or non-spherical

$$\frac{A}{V} = \frac{K_S}{D} \text{ or } K_s = \frac{AD}{V}$$

Where K_s = shape factor; D = the diameter of a sphere of equivalent volume as the nonspherical particle.

continued

Power density

$$PD = \frac{P}{A}$$

where PD = power density, W/mm^2; P = power entering the surface, W; and A= surface area over which the energy is entering, mm^2.

Heat required to melt a given volume of metal

$$U_m = KT_m^2$$

where U_m = the unit energy for melting, J/mm^3; T_m = melting point of the metal on an absolute temperature scale, °K(°R); and K = constant whose value is 3.33 x 10^{-6} JK^{-2}mm^{-3}, when the Kelvin scale is used (and K = 1.467 x 10^{-5} JR^{-2}mm^{-3} for the Rankine temperature scale).

Heat energy available for welding:

$$H_w = f_1 f_2 H$$

where H_w = net heat available for welding, J, f_1 = heat transfer factor, f_2 = the melting factor, and H = the total heat generated by the welding process, J.

Energy balance equation

$$H_w = U_m V$$

Where H_w = net heat energy used by the welding operation, J; U_m = unit energy required to melt the metal, J/mm^3; and V = the volume of metal melted, mm^3.

Rate balance equation

$$R_{Hw} = U_m R_{WV}$$

Where R_{Hw} = rate of heat energy delivered to the operation for welding, J/s = W; and R_{WV} = volume rate of metal welded, mm^3/s.

$$R_{Hw} = f_1 f_2 R_H = U_m A_w v$$

here f_1 and f_2 are the heat transfer and melting factors; R_H = rate of input energy generated by the welding power source, W; A_w = weld cross-sectional area, mm^2; and v = the travel velocity of the welding operation, mm/s.

continued

Time to complete a single layer in rapid prototyping

$$T_i = \frac{A_i}{vD} + T_r$$

Where T_i= time to complete layer i; A_i = area of layer i; v = average scanning speed of the laser beam at the surface; D = diameter of the laser beam at the surface, and T_r = repositioning time between layers.

Build cycle time (stereolithography)

$$T_c = \sum_{i=1}^{nl} T_i$$

Where T_c = the STL build cycle time; and nl = the number of layers used to approximate the part.

END